

Exercise Set 4

1) AWS vs Google Cloud vs Azure

A. What services can you use for free?

Amazon Web Services (AWS) - AWS Free Tier includes a mix of “always free” services plus a free plan with credits for new accounts.

Google Cloud (GCP) - Google Cloud offers selected products free within monthly Free Tier limits, plus product-specific trials.

Microsoft Azure (Azure) - Azure provides two types of free services: some free for the first 12 months and many always-free up to monthly limits.

B. How much does it cost to run a VM for one day? (non-free plan)

Example 1: 2 vCPU, 8 GB RAM (general purpose)

Provider	Example VM type	On-demand \$/hour	Cost for 24h
AWS	m6i.large	\$0.096/hr	\$2.304/day
Azure	D2s v5	\$0.096/hr	\$2.304/day
GCP	n2-standard-2	\$0.0971/hr	\$2.3304/day

Example 2: 4 vCPU, 16 GB RAM (general purpose)

Provider	Example VM type	On-demand \$/hour	Cost for 24h
AWS	m6i.xlarge	\$0.192/hr	\$4.608/day
Azure	D4s v5	\$0.192/hr	\$4.608/day
GCP	n2-standard-4	\$0.1942/hr	\$4.6608/day

C. How much does storage cost?

To compare “object storage” across clouds (the closest apples-to-apples): AWS S3 vs Azure Blob vs Google Cloud Storage.

Provider	Storage product	Example price (per GB-month)
AWS	S3 Standard	\$0.023/GB-month (first 50 TB)
Azure	Blob Storage Hot (pay-as-you-go example)	~\$0.0208/GB-month (example shown in Microsoft Learn)
GCP	Cloud Storage Standard	Pricing varies by location class; see GCP pricing pages (regional/dual-region have different rates)

D. Compare prices and expected performance of different VM offerings. Is price linear with performance?

Within the same family and generation, price is often close to linear when you scale vCPU and RAM proportionally:

- 2 vCPU/8 GB to 4 vCPU/16 GB roughly doubles price across AWS, Azure, and GCP in the example

E. Which of the three is best?

A defensible “no single best” answer (typical in coursework):

- AWS tends to win on breadth of service and ecosystem maturity
- Azure is often best if your company is deeply invested in Microsoft enterprise stack (AD, W, V)
- GCP is often strongest for data analytics and ML-heavy teams

F. If your company wants to start using cloud, which would you pick and why?

A solid default choice:

- Pick AWS if you want the widest service catalog, the biggest hiring pool, and lots of references
- Pick Azure if you're Microsoft-heavy (identity, security policies, Windows workloads, M365, etc)
- Pick GCP if your early roadmap is dominated by data/ML pipelines and you want tight integration with other Google services

2) Would cloud computing have taken off like this without virtualization?

Probably not at anything like today's scale.

Why:

- Virtualization (VMs, and later containers) makes multi-tenancy safe enough (isolation), enabling shared infrastructure
- Without it, cloud would look more like "hosted dedicated servers" - slower to provision, higher cost, less flexible

3-5) Armbrust et al. (2010)

1. Availability / Business Continuity - Partially true Cloud availability improved massively (multi-zone, multi-region, managed failover), but major outages still happen and customers still need resilience design.
2. Data Lock-in - Mostly true Containers, Kubernetes, Terraform, and open APIs reduce lock-in for basic compute, but managed services (databases, event systems, IAM, observability) still create strong lock-in - and data egress costs reinforce it.
3. Data Confidentiality and Auditability - Partially true Security tooling is far better (encryption, KMS/HSM, IAM, logging, compliance), yet confidentiality and auditability remain central concerns, especially for regulated industries and sensitive data.
4. Data Transfer Bottlenecks - Partially true Networks are faster, and physical transfer appliances exist, but large dataset movement, cross-region replication, and egress remain bottlenecks (cost and latency).
5. Performance Unpredictability - Partially true Better instance isolation options exist (dedicated hosts, premium SKUs), but shared infrastructure and noisy-neighbor effects haven't disappeared, especially for storage and multi-tenant services.
6. Scalable Storage - Mostly true This is one area where cloud clearly delivered: object storage and managed storage systems scale elastically and are operationally simpler than most on-prem alternatives.
7. Bugs in Large Distributed Systems - Mostly true Still a fundamental issue. Systems are bigger and more complex; the industry improved practices (SRE, chaos engineering), but distributed failure modes remain hard.
8. Scaling Quickly - Mostly true Elasticity is real, and automation is strong (autoscaling, IaC), but organizational bottlenecks (quotas, architecture, cost controls, governance) often limit "instant scaling" in real companies.
9. Reputation Fate Sharing - Partially true Clouds are better isolated than before, but shared dependencies still bite (shared identity providers, common SaaS outages, shared IP reputation, misconfigurations).
10. Software Licensing - Partially true A lot moved to subscription and consumption pricing, but licensing is still messy (BYOL rules, core-based licensing, marketplace images, audit risk).